Multi-Objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents

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***Abstract*—Automated forensic analysis of identity documents requires simultaneously detecting whether a document has been tampered with (document-level classification) and identifying which regions were altered (pixel-level segmentation). These two objectives are structurally coupled but exist in tension: optimizing one with a fixed loss weight may degrade the other. Existing approaches either treat the two tasks independently or collapse them into a single scalar objective with a manually chosen weight, an arbitrary decision that cannot be justified without exhaustive prior experimentation.**

We propose DocVerify, a multi-task convolutional network paired with a multi-objective hyperparameter optimization (HPO) framework that treats detection quality (PR-AUC) and localization quality (Dice coefficient) as simultaneous objectives. Using the Multiobjective Tree-structured Parzen Estimator (MOTPE) within a 10-fold nested cross-validation protocol (50 trials per fold, 500 model evaluations in total), DocVerify selects hyperparameter configurations from the Pareto-optimal front by minimizing the Euclidean distance to the ideal point $(1,\ 1)$, a criterion that assigns equal importance to detection and localization without any manual weighting.

Evaluated on the FantasyID benchmark, DocVerify achieves PR-AUC $= 0.9921 \pm 0.0058$ and Dice $= 0.856 \pm 0.030$ under nested cross-validation, and a detection F1@0.5 of 0.969 ± 0.014 with localization F1 of 0.807 ± 0.096 on the blind test set (30 seeds). Without any external pre-training, these results are competitive with the top-3 submissions to the DeepID 2025 Challenge at ICCV, training exclusively on FantasyID. An ablation study confirms that Pareto-optimized loss weighting provides a statistically significant improvement in Dice over equal-weight joint training ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent detection accuracy.

***Index Terms*—identity document forensics, forgery detection, forgery localization, multi-task learning, multi-objective hyperparameter optimization, Pareto front, nested cross-validation, U-Net, MOTPE**

IDENTITY document fraud poses a growing threat to national security, financial integrity, and border control worldwide. The widespread availability of image-editing software has made it possible to produce convincing forgeries of passports, national identity cards, and driver's licenses at low cost, enabling identity theft, illegal immigration, and financial crime [?]. Automated forensic analysis of identity documents is therefore a critical component of modern digital identity verification pipelines.

Detection of forged identity documents is inherently a two-level problem. At the *document level*, a system must decide whether a presented document is genuine or has been tampered with (binary detection). At the *pixel level*, it must also identify and delineate which specific regions have been altered (forgery localization). These two objectives are complementary and both are operationally necessary.

Binary detection alone is insufficient for real-world forensic deployments for two reasons. First, a positive detection decision provides no *actionable information* to the investigator: it does not reveal whether the fraudster altered the holder's portrait photograph, falsified textual fields such as name, date of birth, or document number, or combined multiple manipulations. Without spatial evidence, a human expert cannot determine the nature of the fraud, reconstruct the original document, or build a legal case. Second, detection systems are increasingly subject to *explainability requirements* under emerging AI regulations [?]: a binary "tampered" label cannot satisfy the obligation to provide a meaningful explanation of the automated decision. Conversely, localization alone is insufficient as a security gate because producing a pixel-level mask does not yield a reliable document-level confidence score suitable for automated accept/reject decisions. A practical forensic system must address both tasks simultaneously.

Challenges

The primary technical difficulty in joint detection and localization is that the two objectives are in *structural tension*: the classification head benefits from global image features that aggregate evidence across the entire document, while the segmentation head requires spatially precise, pixel-level representations to delineate tampered regions. Prior work has typically addressed this tension by either (i) treating the two tasks independently with separate models, which loses the opportunity for mutual regularization [?], or (ii) collapsing both losses into a single scalar objective via a weighted sum, which requires manually tuning the weight λ before any training begins [?], [?]. The choice of λ is arbitrary, dataset-dependent, and cannot be justified without exhaustive preliminary experiments.

A second challenge is the risk of over-fitting the model-selection criterion to the training distribution. Standard hyper-

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Source code and experimental artifacts are publicly available at https://github.com/HernanDiaz/IDVerify/.

parameter optimization (HPO) maximizes a single validation metric, ignoring potential degradation on the complementary objective. When the HPO criterion is PR-AUC, the model may sacrifice segmentation quality to maximize detection confidence; when it is Dice coefficient, the classification head may be under-trained. Neither outcome is acceptable for a forensic system that must meet simultaneous performance requirements on both tracks.

Contributions

This paper introduces DocVerify, a multi-task forensic document analysis system that addresses both challenges through a principled multi-objective hyperparameter optimization framework. Our contributions are as follows:

1) *Multi-objective Pareto HPO for multitask forensics.* We propose a hyperparameter search framework in which both PR-AUC (detection) and Dice coefficient (localization) are treated as simultaneous optimization objectives. The optimal configuration is selected from the Pareto front of non-dominated solutions by minimizing the Euclidean distance to the ideal point $(1,\ 1)$, a criterion that assigns equal weight to both tasks and requires no manual tuning of task trade-off weights.
2) *DocVerify architecture.* We design an end-to-end multi-task network comprising a lightweight convolutional encoder (six stages, 8–256 channels) coupled to a U-Net decoder with skip connections for pixel-level mask prediction and a dedicated classification head for document-level detection. The architecture is trained jointly with a combined BCE+Dice segmentation loss and a BCE classification loss, with the loss weight λ_{mask} included in the HPO search space.
3) *Rigorous nested cross-validation protocol.* We evaluate DocVerify under a 10-fold outer $\times$ 5-fold inner nested cross-validation scheme, running 50 Optuna MOTPE trials per outer fold. Each trial trains one model per inner fold, yielding 2 500 model evaluations in total, all conducted entirely on the training data. The final performance is estimated on a blind holdout using 30 independent random seeds per variant, and statistical comparisons use the Wilcoxon signed-rank test with Holm–Bonferroni correction.
4) *Competitive results on the DeepID 2025 Challenge benchmark.* Training exclusively on the FantasyID dataset [?] and without any external pre-training, DocVerify achieves a detection F1@0.5 of 0.969 ± 0.014 (Track 1) and a localization F1 of 0.807 ± 0.096 (Track 2), surpassing all FantasyID-only baselines and exceeding the challenge winner on Track 2 (evaluated on an internal holdout; see Section IV for the caveat on comparability).

The source code, training scripts, and experimental artifacts are publicly available at https://github.com/HernanDiaz/IDVerify/.

The remainder of the paper is organized as follows. Section I reviews related work on document forgery detection, multi-task learning, and HPO. Section II describes the DocVerify architecture, loss functions, and optimization framework. Section III presents experimental results including ablation studies and comparison with challenge participants. Section IV discusses the findings, limitations, and implications. Section V concludes the paper.

I. Related Work

A. Document and Image Forgery Detection

Early work on image forensics relied on hand-crafted features such as noise inconsistencies, JPEG compression artifacts, and chromatic aberration patterns to detect copy-move and splicing manipulations [?]. Deep learning substantially shifted the field: convolutional neural networks can learn to detect subtle statistical fingerprints left by image-processing operations, often outperforming engineered pipelines on standard benchmarks [?], [?].

A complementary research thread exploits camera-model fingerprints embedded in image noise. Noiseprint [?] trains a CNN via Siamese contrastive learning to extract camera-model-specific noise residuals; forgeries appear as anomalies in this residual and can be localized without semantic understanding. TruFor [?] generalizes this idea with a transformer fusion network (NoisePrint++) that jointly models RGB semantics and noise-pattern anomalies, achieving strong performance on general image forensics benchmarks with over 876 000 pre-training images. General-purpose detectors such as MMFusion [?], which combines complementary forensic filters (SRM, Bayar convolution) through an early-and-late fusion decoder, and FatFormer [?], which adapts a CLIP-ViT backbone with forgery-aware spatial and frequency adapters, further advance the state of the art for natural-image manipulation. UniFD [?] demonstrates that nearest-neighbor classification in a frozen CLIP feature space, without any discriminative fine-tuning, already achieves competitive detection rates across diverse generative models.

Identity document forgery, however, introduces unique challenges compared to natural-image manipulation: (i) the forged regions are often small (a single text field or a face) relative to the full document, (ii) the background texture is highly structured (guilloché patterns, UV-reactive ink), which creates strong false negatives for general-purpose noise detectors, and (iii) the diversity of document types across countries and issuers demands generalization beyond what natural-image pre-training provides. The SIDTD dataset [?] provides a first large-scale benchmark for this domain and shows that models trained on one set of nationalities degrade severely on unseen ones, motivating few-shot and domain-generalization approaches. The FakeIDet line of work [?] further explores patch-based DINOv2 [?] representations with multi-head self-attention fusion for privacy-preserving fake-ID detection. The DeepID 2025 Challenge [?] was specifically designed to benchmark progress on joint detection and localization using the FantasyID dataset [?], which covers face-swap and text-inpainting attacks on fantasy identity documents. Official evaluations on FantasyID show that the general-purpose baselines (TruFor, MMFusion, FatFormer, UniFD) all achieve false-negative rates above 50% at a 10% false-positive rate threshold [?], confirming that document-agnostic forensic models require domain-specific adaptation.

Among systems trained exclusively on FantasyID, George and Marcel [?] propose EdgeDoc, a lightweight hybrid CNN-Transformer encoder (EdgeNeXt [?]) that takes both the green channel of the document image and a NoisePrint residual as input, feeding a U-Net decoder optimized jointly for classification and pixel-level localization. Naseeb *et al.* [?] propose TwoHead-SwinFPN, a Swin Transformer backbone with a feature pyramid network decoder, CBAM attention modules, and dual-output heads for joint classification and segmentation, reporting 88.61% F1 and 57.24% Dice on FantasyID. Yu *et al.* [?] introduce Re-MTKD, a reinforcement-learning multi-teacher knowledge distillation framework using a ConvNeXt v2 encoder and UPerNet decoder that leverages specialist teachers for copy-move, splicing, and inpainting; the winning Sunlight team adapted this framework to identity documents using over 60 000 additional external training images, achieving the best results in both tracks of the DeepID 2025 Challenge.

DocVerify differs from all prior document-forensics systems in three respects: (i) it systematically searches the task-weight λ_{mask} via multi-objective HPO rather than fixing it heuristically; (ii) it selects the final configuration via a Pareto-optimal criterion that treats detection and localization as explicitly competing objectives; and (iii) it reports uncertainty through nested cross-validation, providing statistically rigorous performance estimates rather than single-split results.

B. Multi-Task Learning for Detection and Localization

Joint detection and localization is a canonical instance of multi-task learning (MTL) [?], [?]. In MTL, shared representations learned for one task regularize learning for related tasks, typically yielding better generalization than independent training when tasks are correlated [?].

The dominant approach to combining task-specific losses is scalar scalarization [?]: a single objective $\mathcal{L} = \sum_t \lambda_t \mathcal{L}_t$ is minimized, with fixed weights λ_t set before training. Kendall *et al.* [?] proposed learning λ_t jointly with the model weights via homoscedastic uncertainty, but this requires an additional learning-rate schedule and is sensitive to initialization. Liu *et al.* [?] introduced gradient-normalization to dynamically balance task gradients, at the cost of additional computation per step. All scalar approaches share a common limitation: the optimal trade-off between tasks is not established before training, yet the weights implicitly commit to a fixed trade-off.

In the image forensics domain, U-Net [?] architectures have been widely adopted for pixel-level localization because their skip connections preserve spatial resolution while the encoder captures high-level semantic anomaly features. Several works combine classification and segmentation branches in a single network, but typically fix the task weights empirically without systematic optimization [?]. Our work differs by including the task weight λ_{mask} in the HPO search space and selecting it via a multi-objective criterion rather than a grid search.

C. Hyperparameter Optimization and Multi-Objective Search

Hyperparameter optimization (HPO) has evolved from manual search and grid search to sequential model-based methods. Tree-structured Parzen Estimator (TPE) [?] models the distribution of good and bad configurations separately and selects candidates that maximize the ratio of the two models, a strategy significantly more sample-efficient than random search [?]. The Optuna framework [?] implements TPE as its default sampler and has become a standard tool for HPO in deep learning.

Multi-objective HPO extends single-objective search to the case where multiple metrics must be jointly optimized. The output is a Pareto-optimal set of configurations rather than a single best point, making the trade-off between objectives explicit [?]. Evolutionary algorithms such as NSGA-II [?] are the most established approach, but they require large population sizes to maintain Pareto-front diversity. The Multiobjective TPE (MOTPE) sampler [?] adapts the TPE surrogate model to the multi-objective setting, achieving competitive Pareto-front coverage with far fewer function evaluations than evolutionary methods—a critical advantage when each evaluation requires training a deep neural network.

To our knowledge, this is the first work to apply Pareto-based multi-objective HPO to the problem of joint identity-document forgery detection and localization, and the first to use the resulting Pareto front as the primary model-selection criterion in a nested cross-validation framework.

II. Methodology

A. Dataset

We train and evaluate DocVerify exclusively on FantasyID [?], a publicly released benchmark for identity document forgery detection and localization developed by Idiap Research Institute for the DeepID 2025 Challenge [?]. FantasyID comprises approximately 3 284 labeled images of synthetic fantasy identity documents in two classes: *bonafide* (genuine, unmodified) and *attack* (tampered). Attack images are produced via two representative manipulation types: (i) *face swap*, replacing the portrait photograph with a different identity, and (ii) *text inpainting*, overwriting text fields (name, date of birth, document number) with forged content.

Each image is accompanied by a JSON annotation file specifying bounding-box regions with a `region_provenance: altered` field that identifies the tampered areas. We convert these annotations to binary segmentation masks by rasterizing the altered bounding boxes at the training resolution (224×224 pixels). Bonafide images have all-zero masks. We combine the FantasyID `train` and `test` partitions into a single pool of 3 284 images and apply a `GroupShuffleSplit` (grouping by document identity, test_size $= 0.15$, `random_state`=42) to obtain approximately 2 791 development images and 493 blind holdout images, ensuring that all captures of the same document identity remain in the same partition and preventing any form of data leakage between splits. The holdout set is accessed only for the final blind evaluation (Section III-C2).

B. Architecture

DocVerify is a multi-task encoder-decoder network designed for simultaneous document-level classification and pixel-level forgery localization. Figure 1 shows the overall structure.

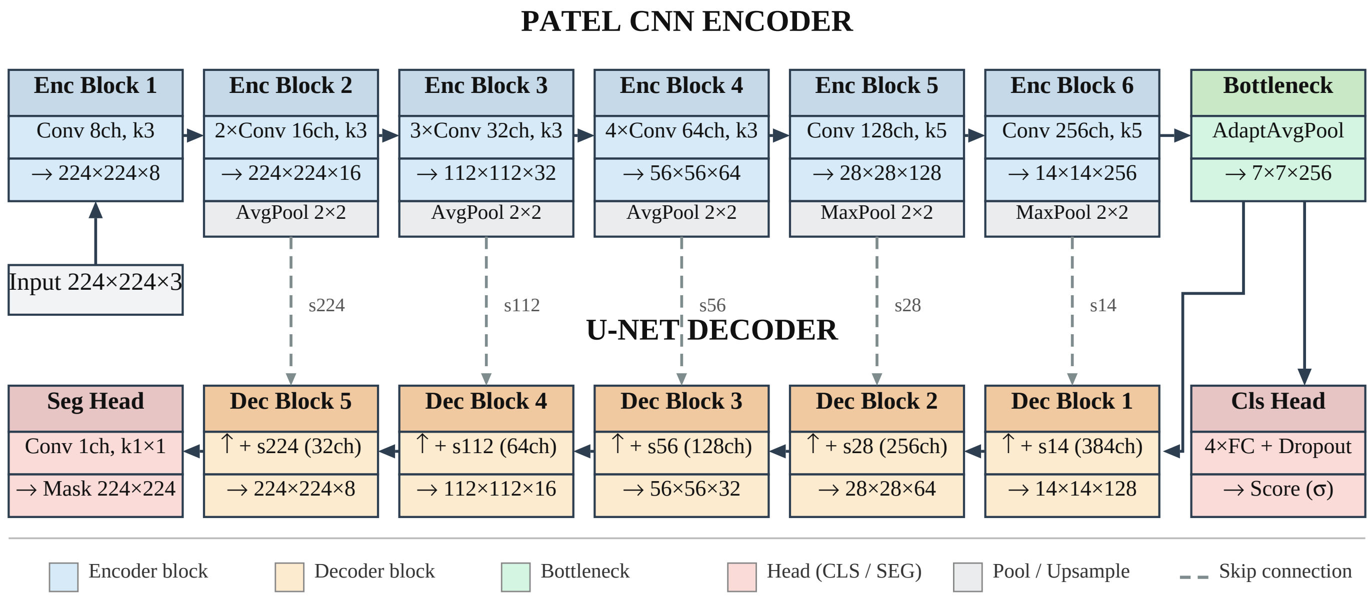


Fig. 1. DocVerify architecture. The encoder maps a $3\times224\times224$ input to a $256\times7\times7$ bottleneck through six convolutional blocks ($8\to256$ channels). The U-Net decoder reconstructs the forgery mask via skip connections; the classification head outputs a document-level tampered/genuine score.

1) Encoder: The encoder is a lightweight six-block convolutional backbone inspired by the dense CNN (D-CNN) architecture proposed by Patel et al. [?] for deepfake image detection, adapting its increasing channel-width progression and mixed pooling strategy to the document-forensics domain. It employs an increasing channel progression of $8 \to 16 \to 32 \to 64 \to 128 \to 256$ filters. All convolutional layers use LeakyReLU activations (negative slope $\alpha = 0.2$) [?] and batch normalization [?]. Blocks 1–4 use 3×3 convolutions; blocks 5 and 6 use 5×5 convolutions to capture wider spatial context at lower feature-map resolutions. Spatial downsampling is applied after each block: average pooling after blocks 2 and 3 (to preserve boundary information for localization), and max pooling after blocks 4 and 5 (to achieve stronger translation invariance for classification). The encoder maps a $3\times224\times224$ input to a bottleneck of $256 \times 7 \times 7$.

Skip connections are extracted from each block prior to pooling, yielding feature maps at five resolutions: 224×224 (16 ch), 112×112 (32 ch), 56×56 (64 ch), 28×28 (128 ch), and 14×14 (256 ch).

2) Classification Head: The classification head applies global average pooling to the 7×7 bottleneck, yielding a 256-dimensional feature vector. This is passed through four fully-connected layers ($256 \to 32 \to 16 \to 16 \to 1$) interleaved with dropout and LeakyReLU activations, producing a single classification logit $\hat{y}_{\text{cls}}$.

3) Segmentation Head (U-Net Decoder): The segmentation head reconstructs the spatial mask through a five-stage U-Net decoder [?]. A 1×1 projection convolution first maps the bottleneck to C_{dec} channels (a tunable hyperparameter, see Section II-D). Each decoder block bilinearly upsamples by a factor of two, concatenates the corresponding encoder skip connection, and applies two 3×3 convolutions with LeakyReLU activations—mirroring the standard U-Net design. The channel width halves at each stage from C_{dec} down to $C_{\text{dec}}/16$. A final 1×1 convolution produces the segmentation logit $\hat{Y}_{\text{mask}}$ at the original input resolution.

Both output logits are kept raw (no sigmoid at inference time); sigmoid is applied numerically within the loss functions for numerical stability.

C. Loss Functions

The combined training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{mask}}\, \mathcal{L}_{\text{seg}}, \tag{1}$$

where $\lambda_{\text{mask}} \in [0.5, 3.0]$ is a tunable weight included in the HPO search space (Section II-D).

a) Classification loss.: $\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\hat{y}_{\text{cls}}, y_{\text{cls}})$, where $y_{\text{cls}} \in \{0, 1\}$ is the document-level label.

b) Segmentation loss.: $\mathcal{L}_{\text{seg}} = \text{BCEWithLogits}(\hat{Y}_{\text{mask}}, Y_{\text{mask}}) + \mathcal{L}_{\text{Dice}}(\sigma(\hat{Y}_{\text{mask}}), Y_{\text{mask}})$, where $\sigma(\cdot)$ denotes the sigmoid function and the Dice loss [?] is computed per image and averaged over the batch:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad \epsilon = 10^{-6}. \tag{2}$$

The BCE–Dice combination was introduced for volumetric medical image segmentation [?] and has since become standard practice for pixel-level tasks with severe class imbalance [?]; it directly addresses the foreground/background imbalance caused by small forgery regions such as a single altered text field.

c) Training details.: All models are trained with AdamW [?] and ReduceLROnPlateau (factor 0.5, patience 3). Gradient clipping with a maximum ℓ_2 norm of 1.0 is applied at every step. Mixed-precision training (FP16 forward pass, FP32 gradient accumulation via the PyTorch AMP GradScaler [?], [?]) is used throughout, reducing memory usage without affecting convergence. Input images are normalized to $[0, 1]$ and resized to 224×224 pixels.

D. Multi-Objective HPO

Our HPO formulation treats detection quality and localization quality as two independent objectives to be simultaneously maximized:

$$\max_{\boldsymbol{\theta}\in\Theta} \left(\text{PR-AUC}(\boldsymbol{\theta}),\ \text{Dice}(\boldsymbol{\theta})\right), \tag{3}$$

where $\boldsymbol{\theta}$ is a hyperparameter configuration and Θ is the search space defined in Table I.

The LeakyReLU slope α was initially included in the search space but removed after a preliminary study of 250 trials showed no signal on validation performance across the range $[0.1, 0.4]$; it is fixed at $\alpha = 0.2$ throughout.

1) MOTPE Sampler: We use the Multiobjective Tree-structured Parzen Estimator (MOTPE) [?] as implemented in Optuna [?] with NSGA-II [?] as a fallback sampler. MOTPE extends the single-objective TPE by modeling the joint distribution of hyperparameter configurations in the objective space, efficiently focusing the search toward the Pareto-optimal frontier without requiring an explicit scalarization function. This is key for our setting, where neither objective should be privileged a priori.

2) Pareto-Based Model Selection: After running 50 Optuna trials on the inner validation split of each outer fold, we identify the set of Pareto-optimal trials (non-dominated solutions). From this set, we select the configuration $\boldsymbol{\theta}^*$ that minimizes the Euclidean distance to the ideal point $(1, 1)$ in the objective space:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\mathcal{P}} \sqrt{(1-\text{PR-AUC})^2 + (1-\text{Dice})^2}, \tag{4}$$

where $\mathcal{P}$ denotes the Pareto front. This criterion requires no user-specified weighting and is consistent with the DeepID 2025 Challenge evaluation protocol, which ranks systems independently on Track 1 (detection) and Track 2 (localization), assigning equal importance to both.

Figure 2 illustrates the objective space across all 500 HPO trials (10 folds $\times$ 50 trials), with the Pareto-optimal trials highlighted.

E. Nested Cross-Validation

To obtain unbiased performance estimates while keeping HPO blind to the test data, we adopt a three-level nested cross-validation (NCV) scheme illustrated in Figure 3.

a) Outer loop (10 folds).: The FantasyID training set is partitioned into 10 stratified folds. Each fold serves once as the test set for a held-out performance estimate.

b) Inner loop (5 folds, HPO).: For each outer fold, the remaining 9/10 of the data is further split into 5 inner folds. The 50 Optuna MOTPE trials are run on these inner splits, with each trial training for up to $E_{\text{trial}} = 15$ epochs with patience $= 1$ on the inner validation loss. The optimal configuration $\boldsymbol{\theta}^*$ is selected via (4).

c) Final model training.: The selected $\boldsymbol{\theta}^*$ is used to retrain the model from scratch on all 9/10 outer-fold training data for up to $E_{\text{final}} = 100$ epochs with early stopping (patience $=$ 12, monitoring the distance-to-ideal criterion on a held-out inner validation slice). The trained model is then evaluated on the held-out outer-fold test set to produce one unbiased performance estimate.

This design ensures that neither the HPO process nor the final training sees the test fold, eliminating the optimistic bias that arises when hyperparameter selection and performance estimation share data [?]. The total number of model trainings is $10 \times 5 \times 50 = 2\,500$ (HPO) plus 10 final models.

III. Experiments and Results

A. Experimental Setup

All experiments are conducted on a single NVIDIA RTX 5060 Ti (16 GB VRAM) under Windows 11, using PyTorch 2.10.0 with CUDA 13.0 and Optuna 3.x. The full nested cross-validation pipeline (including 2 500 HPO trials and 10 final models) required approximately 59 hours of wall-clock time. The scalar experiment (Section III-D) required an additional 4.6 hours.

Images are loaded into GPU memory at the start of each outer fold to eliminate CPU-to-GPU transfer overhead during inner-fold HPO trials (per-fold VRAM cache). This optimization reduced per-trial training time from approximately 12 to 3.5 minutes.

B. Nested Cross-Validation Results

Table II reports the per-fold test metrics obtained from the 10 outer folds of the nested cross-validation. Each metric is computed on the held-out test fold using the model trained with the Pareto-selected hyperparameter configuration.

PR-AUC is consistently high across all folds (0.977–0.999, mean 0.9921 ± 0.0058), confirming reliable document-level detection generalization. Dice scores show slightly more fold-to-fold variability (0.800–0.909, mean 0.856 ± 0.030), reflecting the inherent difficulty of pixel-level localization under diverse forgery patterns. No fold exhibits a performance collapse, confirming that the model does not over-fit to any particular subset of the training distribution.

The Pareto-selected λ_{mask} values cluster consistently between 2.3 and 2.7 (eight of ten folds), with folds F5 and F9 selecting lower weights (1.41 and 1.95), which coincide with the two lowest Dice scores in the table. This observation suggests that λ_{mask} plays a decisive role in localization quality and motivates including it in the HPO search space rather than fixing it a priori.

Figure 3 visualizes the per-fold PR-AUC and Dice values with their cross-fold means.

C. Ablation Study

1) Variants: To isolate the contribution of each design decision, we evaluate four model variants on the blind test set using 30 independent random seeds per variant:

- *Multitask* (proposed): joint detection + localization with Pareto-optimized λ_{mask}.
- `cls_only`: classification head only; no segmentation head or segmentation loss.
- `seg_only`: segmentation head only; classification is performed by thresholding the mean mask probability.

TABLE I
HPO SEARCH SPACE

Parameter	Distribution	Range	Scale
η (learning rate)	log-uniform	$[5 \times 10^{-5},\ 9 \times 10^{-4}]$	log
λ_{wd} (weight decay)	log-uniform	$[10^{-7},\ 10^{-4}]$	log
p (dropout rate)	uniform	$[0.1,\ 0.4]$	linear
C_{dec} (decoder channels)	categorical	$\{96, 128, 192, 256\}$	—
λ_{mask} (mask loss weight)	uniform	$[0.5,\ 3.0]$	linear

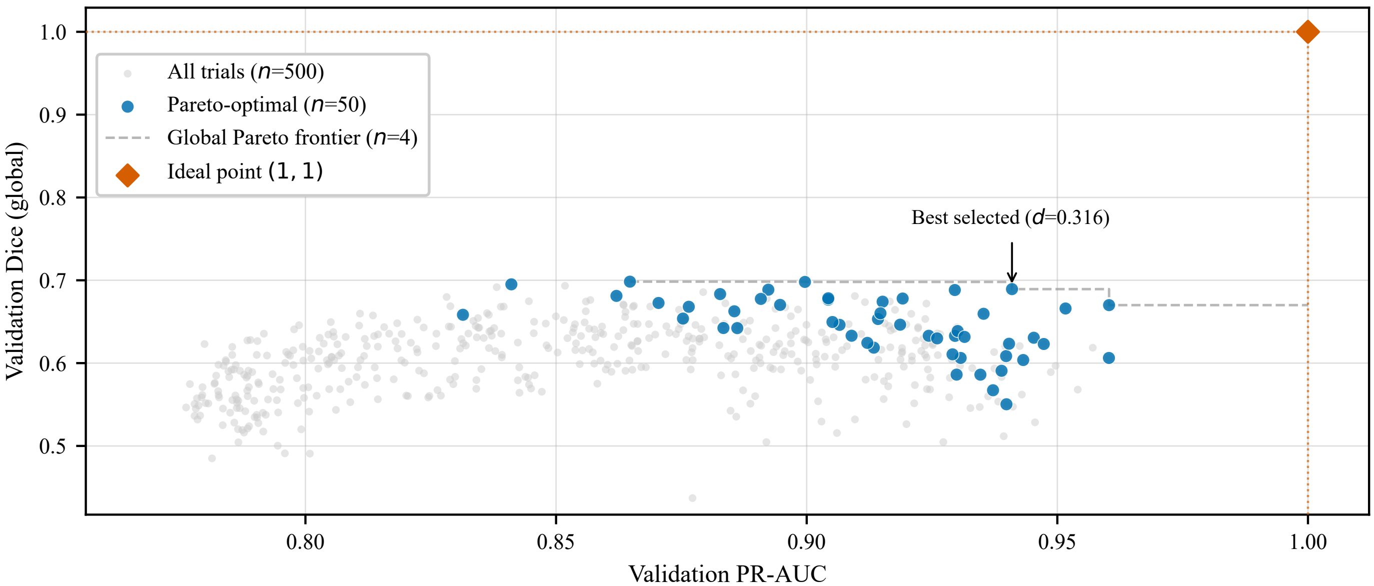


Fig. 2. Validation objective space across all 500 HPO trials (10 outer folds $\times$ 50 Optuna MOTPE trials per fold). Grey points: evaluated configurations; blue points: Pareto-optimal (non-dominated) trials; dashed line: global Pareto frontier. The red diamond at $(1,\ 1)$ marks the ideal point; the annotated point is the selected configuration (minimum Euclidean distance $d = 0.316$).

TABLE II
PER-FOLD TEST RESULTS FROM 10-FOLD NESTED CROSS-VALIDATION. $\overline{w}_{\text{MASK}}$: PARETO-SELECTED MASK LOSS WEIGHT.

Fold	PR-AUC	Dice	ROC-AUC	BACC	$\overline{w}_{\text{mask}}$
F1	0.9939	0.861	0.9848	0.9491	2.46
F2	0.9769	0.871	0.9615	0.9319	2.70
F3	0.9993	0.909	0.9981	0.9470	2.47
F4	0.9915	0.858	0.9773	0.8986	2.43
F5	0.9948	0.827	0.9869	0.9318	1.41
F6	0.9936	0.874	0.9830	0.9319	2.89
F7	0.9940	0.861	0.9834	0.9479	2.67
F8	0.9967	0.887	0.9911	0.9557	2.29
F9	0.9896	0.807	0.9722	0.9023	1.95
F10	0.9909	0.800	0.9765	0.9294	2.65
Mean	**0.9921**	**0.856**	**0.9815**	**0.9326**	**2.39**
Std	**0.0058**	**0.030**	**0.0110**	**0.0192**	**0.43**

- `unweighted_losses`: full multitask architecture with equal task weights ($\lambda_{\text{cls}} = \lambda_{\text{mask}} = 0.5$), bypassing HPO entirely.

2) Results: Table III reports mean and standard deviation over 30 seeds for PR-AUC and Dice, along with Wilcoxon signed-rank tests with Holm–Bonferroni correction comparing each variant to the Multitask baseline. Effect sizes are reported as Cohen's d [**?**].

The results confirm three design choices. First, the joint multitask objective is essential: `cls_only` produces near-zero Dice (0.002 ± 0.008), confirming that the segmentation head cannot be removed without catastrophic localization failure. Second, a dedicated classification head is necessary: `seg_only` achieves competitive Dice (0.867) but its PR-AUC drops to 0.728 ± 0.053, demonstrating that the segmentation logit alone is an insufficient detection signal. Third, the Pareto-optimized loss weights provide a measurable benefit over equal weights: `unweighted_losses` matches Multitask on PR-AUC ($p = 0.083$, not significant) but shows significantly lower Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect), validating the

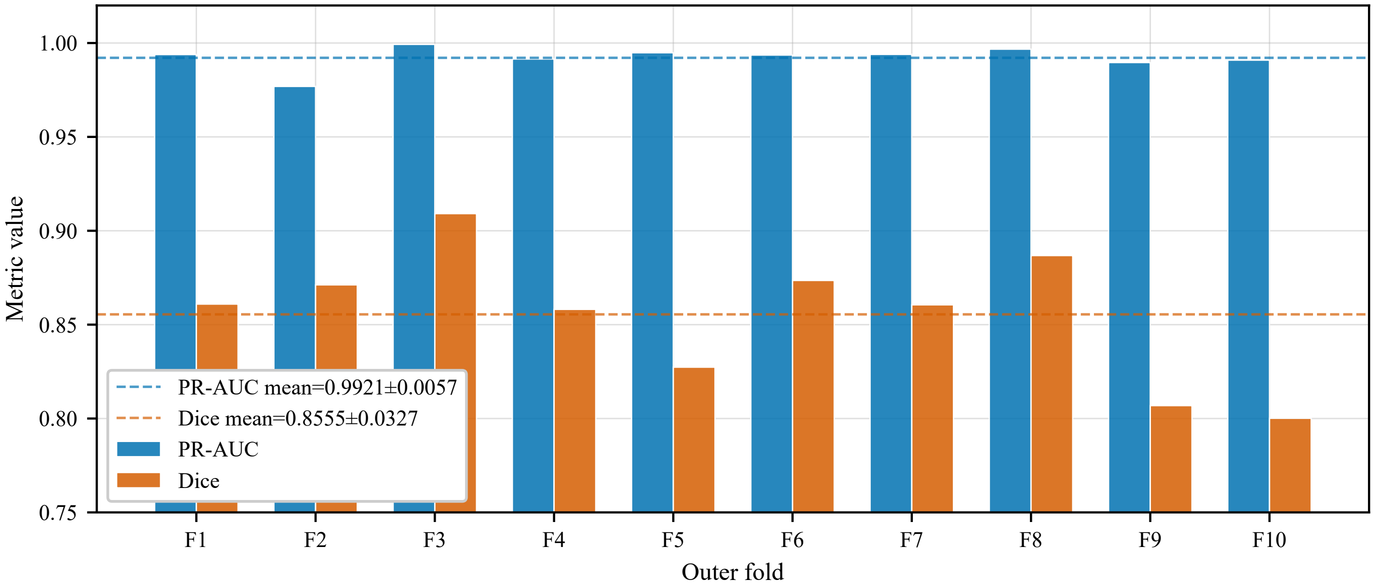


Fig. 3. Per-fold test metrics from 10-fold nested cross-validation. Blue bars: PR-AUC; orange bars: Dice. Dashed lines indicate cross-fold means (PR-AUC $= 0.9921 \pm 0.0060$; Dice $= 0.8555 \pm 0.0345$). The y-axis is truncated at 0.75 to emphasize fold-to-fold variation.

TABLE III
ABLATION STUDY RESULTS (BLIND TEST, 30 SEEDS). p: WILCOXON SIGNED-RANK WITH HOLM CORRECTION; d: COHEN'S d. SIGNIFICANCE: *** $p < 0.001$; NS $p \geq 0.05$.

	PR-AUC		Dice			
Variant	Mean ± Std	vs. Multitask	Mean ± Std	vs. Multitask	p_{holm}	d
Multitask (ours)	$\mathbf{0.9967 \pm 0.0021}$	—	$\mathbf{0.875 \pm 0.018}$	—	—	—
cls_only	0.9503 ± 0.0744	***	0.002 ± 0.008	***	1.3×10^{-8}	large
seg_only	0.7277 ± 0.0528	***	0.867 ± 0.023	ns	0.288	small
unweighted_losses	0.9958 ± 0.0016	ns	0.857 ± 0.016	***	1.3×10^{-4}	1.01

contribution of the HPO weighting scheme specifically for localization quality.

Figure 4 shows the distribution of scores across 30 seeds as violin plots.

D. Pareto HPO vs. Scalar Baselines

To further contextualize the multi-objective HPO, we compare Pareto-based selection against two scalar alternatives that select a configuration from the same 50 MOTPE trials by maximizing a single inner-validation metric: `by_prauc` (maximize PR-AUC) and `by_dice` (maximize Dice).

No statistically significant differences are found between Pareto and either scalar criterion (all $p \geq 0.05$, Wilcoxon + Holm, $n = 10$). The three methods achieve closely matched absolute performance, with differences below 0.004 on PR-AUC and 0.008 on Dice. Pareto selection shows higher variance in Dice (std 0.035 vs. 0.015–0.017 for scalar methods), which can be attributed to sensitivity to inner-fold composition in folds where the Pareto-optimal λ_{mask} departs from the cluster mode (F5, F9).

The key advantage of Pareto HPO is therefore methodological rather than numerical: it (i) selects configurations without privileging either objective a priori, consistently with the DeepID evaluation protocol; (ii) makes the detection-localization trade-off explicit through the Pareto front (Fig. 2), enabling practitioners to choose alternative operating points; and (iii) requires no post-hoc justification for the selection criterion.

Figure 5 shows the distribution of per-fold scores for all three selection strategies.

E. Comparison with DeepID 2025 Challenge Participants

Table V compares DocVerify against the top participants of the DeepID 2025 Challenge on Track 1 (detection F1 at threshold 0.5) and Track 2 (per-image pixel-level F1). Challenge metrics are computed using our challenge evaluation script over our internal holdout test split (15% of FantasyID) and 30 random seeds.

Track 1 (detection). DocVerify achieves $\text{F1@0.5} = 0.969 \pm 0.014$, surpassing all systems that trained exclusively on FantasyID, including AG/EdgeDoc (+1.1 pp) and UAM-Biometrics (+25.7 pp), and outperforming Incode despite the latter using 100K proprietary images. The gap to Sunlight (−2.2 pp) is attributable to Sunlight's use of over 60K additional external training images and multi-stage pre-training.

Track 2 (localization). DocVerify achieves $\text{F1}_{\text{loc}} = 0.807 \pm 0.096$, which exceeds all four challenge finishers including the overall winner Sunlight (0.784). This is the strongest result reported for localization on this benchmark among systems that trained without external data.

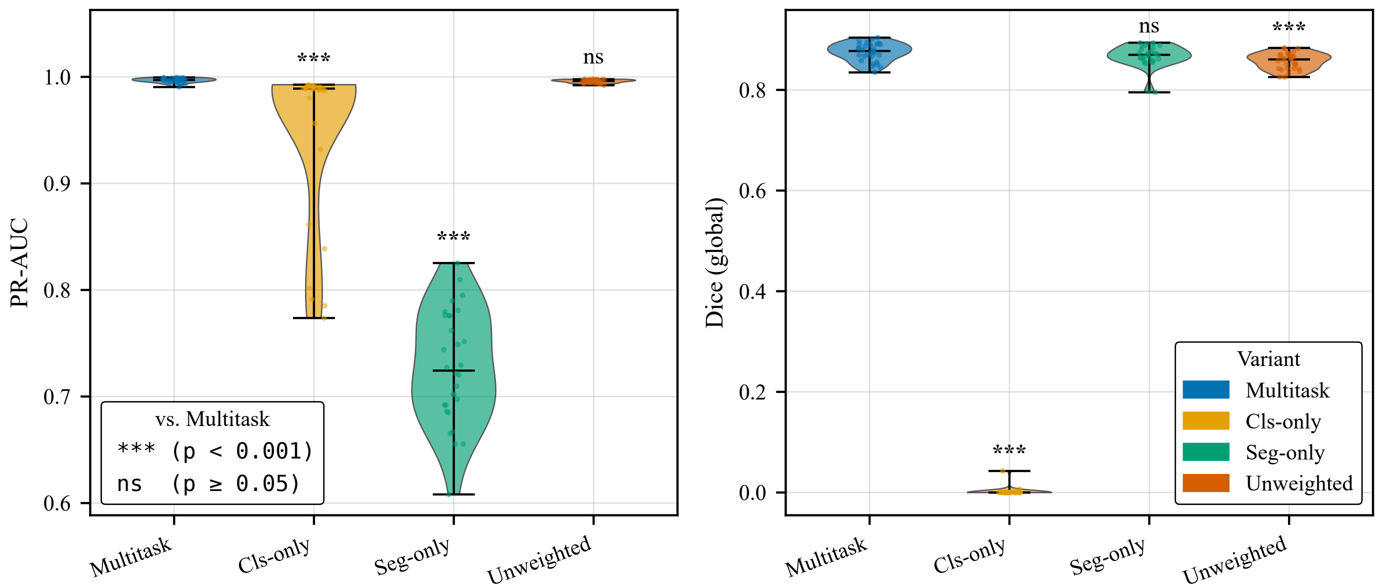


Fig. 4. Ablation study: violin plots of PR-AUC (left) and Dice (right) for four variants over 30 seeds on the blind test set. Significance symbols above each violin denote Wilcoxon + Holm vs. Multitask (*** $p < 0.001$; ns $p \geq 0.05$).

TABLE IV
COMPARISON OF HPO SELECTION CRITERIA ON NESTED CV (10 FOLDS). WILCOXON + HOLM VS. PARETO BASELINE; NS $p \geq 0.05$.

Method	PR-AUC	Dice	p (PR-AUC)	p (Dice)
Pareto (ours)	0.9921 ± 0.0060	0.8555 ± 0.0345	—	—
Scalar by_prauc	0.9958 ± 0.0015	0.8574 ± 0.0166	ns	ns
Scalar by_dice	0.9936 ± 0.0033	0.8630 ± 0.0147	ns	ns

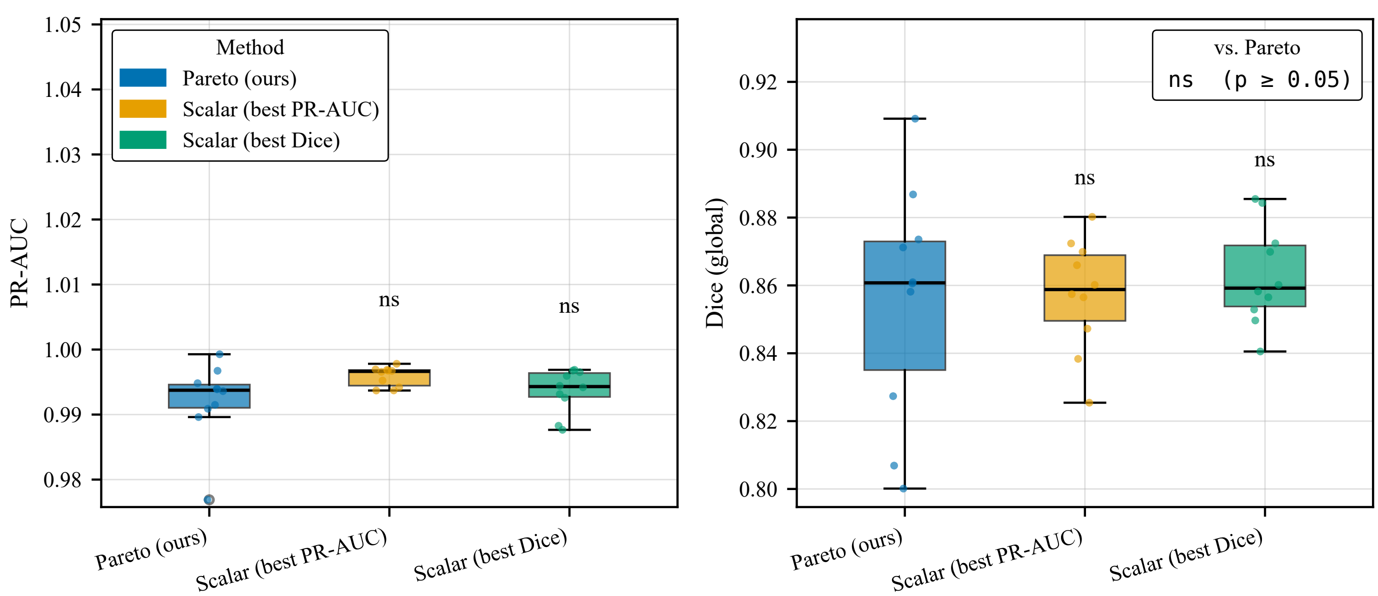


Fig. 5. Pareto HPO vs. scalar selection on nested CV (10 folds). Box plots of PR-AUC (left) and Dice (right); jittered dots show individual fold results. Significance vs. Pareto: ns $p \geq 0.05$ (Wilcoxon + Holm, $n = 10$).

The standard deviation on Track 2 (± 0.096) is notably larger than on Track 1 (± 0.014). Post-hoc analysis reveals that this variance is driven by F1 on bonafide images (0.733 ± 0.181): seeds where the model occasionally misclassifies a bonafide image produce false-positive mask pixels, which the per-image F1 metric penalizes heavily. F1 on attack images is stable (0.881 ± 0.015), confirming that segmentation of forged regions is robust across seeds.

Caveat. DocVerify was evaluated on our internal 15% holdout, whereas challenge systems were evaluated on the

TABLE V
COMPARISON WITH DEEPID 2025 CHALLENGE PARTICIPANTS AND RELATED WORK. †: EVALUATED ON THE OFFICIAL CHALLENGE TEST SET; ‡: REPORTED ON FANTASYID VALIDATION SET (DICE COEFFICIENT); DOCVERIFY EVALUATED ON AN INTERNAL 15% HOLDOUT (SEE TEXT). "—" INDICATES METRIC NOT REPORTED OR TRACK NOT ENTERED.

System	F1 det (thr=0.5)	F1 loc (per-image)	Training Data
Sunlight [?] (1st)†	0.991	0.784	60K+ external images
Incode (2nd)†	0.868	—	100K proprietary images
AG / EdgeDoc [?] (3rd)†	0.958	0.686	FantasyID only
UAM-Biometrics (4th)†	0.712	0.620	FantasyID only
TruFor [?] baseline†	0.807	0.590	876K pre-training images
TwoHead-SwinFPN [?]‡	—	0.572	FantasyID only
DocVerify (ours)	$\mathbf{0.969 \pm 0.014}$	$\mathbf{0.807 \pm 0.096}$	FantasyID only

official test set released by the organizers. The two evaluation sets are not identical, and the comparison is therefore indicative rather than strictly controlled. This limitation is discussed further in Section IV.

The results are summarized in Table V above.

F. Qualitative Analysis

Figure 6 shows representative outputs for a bonafide document and a forged document with five altered regions. For the bonafide sample the model correctly returns an empty mask, producing no false alarms. For the attack sample the predicted mask successfully localises all five tampered fields — photograph, given name, date of birth, document number, and expiry date — and the red overlay confirms tight spatial alignment between the predicted regions and the visible ground-truth alterations.

IV. DISCUSSION

A. Why Multitask Outperforms cls_only on Localization

The most striking result in the ablation study is the collapse of Dice to near zero when the segmentation head is removed (`cls_only`: 0.002 ± 0.008 vs. 0.875 ± 0.018 for Multitask, $p = 1.3 \times 10^{-8}$). This is expected by construction: without a segmentation head, no pixel-level predictions are produced. However, the reciprocal comparison reveals a more nuanced picture: `seg_only` achieves a Dice score of 0.867 ± 0.023 that is not significantly lower than Multitask ($p = 0.288$), yet its PR-AUC drops to 0.728 ± 0.053 ($p = 1.5 \times 10^{-8}$). This asymmetry reveals that *it is the classification task that benefits most strongly from joint training*: the shared encoder, regularized by the segmentation loss, learns spatially-aware features that improve document-level detection far beyond what a classification-only or threshold-on-mask approach achieves. Conversely, the segmentation head performs almost equally well whether or not a dedicated classification head is present, confirming that the U-Net decoder captures sufficient forgery evidence on its own. The binary detection of an identity document forgery requires aggregated global evidence that the segmentation branch, trained to produce spatial maps, cannot reliably provide without the explicit classification objective.

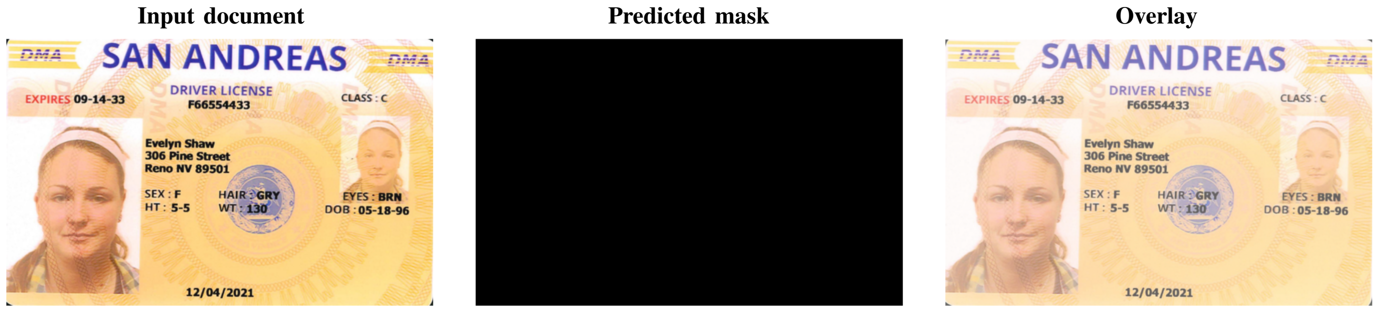
The comparison between Multitask and `unweighted_losses` isolates the contribution of HPO loss weighting. These two variants share the same architecture and joint training objective; they differ only in whether λ_{mask} is set by Pareto HPO (mean 2.39 ± 0.43 across folds) or fixed at 0.5. The Pareto-optimized weights yield a significantly higher Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent PR-AUC ($p = 0.083$). This finding confirms that the loss weighting is the principal driver of localization quality and that a value near $\lambda_{\text{mask}} \approx 2.4$ is broadly optimal for this dataset.

B. On the Pareto Advantage

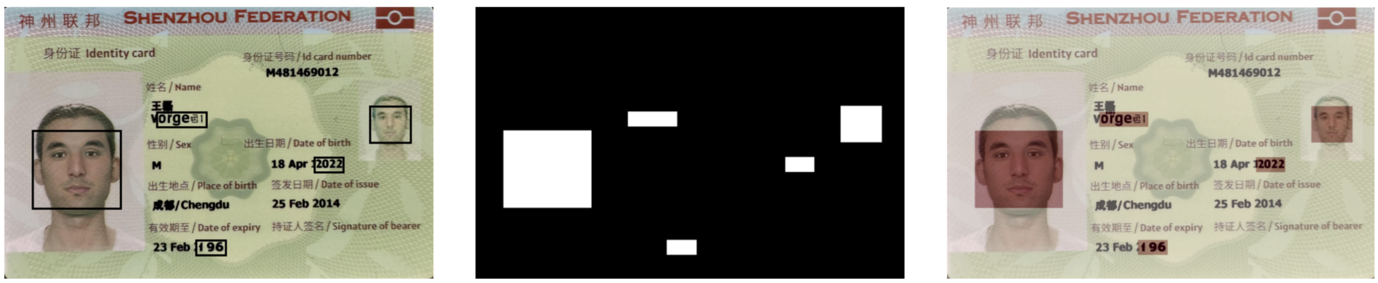
The scalar scalarization experiment (Section III-D) shows that Pareto HPO does not outperform scalar selection in absolute metric terms on this dataset ($n = 10$ folds, no significant differences). This result should be interpreted carefully. The primary claim of the paper is not that Pareto HPO produces higher numbers than scalar HPO, but that it provides a principled, objective-agnostic selection mechanism that:

1) *Eliminates the manual weight choice.* Scalar selection requires the practitioner to declare a single performance criterion (PR-AUC or Dice) as the optimization target. In a forensic deployment context, neither is clearly dominant: detection accuracy determines the security gate reliability, while localization accuracy determines the quality of forensic evidence. Pareto HPO avoids this choice.
2) *Reveals the trade-off landscape.* The 500-point objective space in Figure 2 provides a global picture of how PR-AUC and Dice interact across hyperparameter configurations. In particular, it shows that high-Dice configurations (≥ 0.70) consistently achieve PR-AUC ≥ 0.90, confirming that the two objectives are broadly complementary (not conflicting) in this setting. This information is invisible to scalar HPO.
3) *Is consistent with the evaluation protocol.* The DeepID 2025 Challenge ranks systems independently on Track 1 and Track 2, assigning no implicit weighting. Minimizing the distance to the ideal point (Eq. 4) mirrors this evaluation semantics.

Numerically, the significant difference against `unweighted_losses` in the blind test ($d = 1.01$ on Dice) provides the empirical validation that HPO-based weighting matters. The scalar vs. Pareto comparison is best understood as demonstrating that, given a well-tuned λ_{mask},



(a) Bonafide document — no alterations detected



(b) Forged document — five altered regions localised

Fig. 6. Qualitative results on two FantasyID samples. **Left:** input document (forged regions framed in the attack sample). **Centre:** binary mask predicted by DocVerify. **Right:** input document with predicted mask overlaid in red.

the specific selection criterion (distance to ideal point vs. maximizing a single metric) has a second-order effect.

C. Limitations and Future Work

a) Single dataset and limited cross-domain generalization.: All results are obtained on FantasyID. While the dataset covers multiple document types and two attack categories, it is a synthetic dataset generated at a single research institute. Generalization to real-world identity documents, unseen document designs, or attacks not present in FantasyID (e.g., physical tampering, scanner artifacts) remains an open question and a known limitation of all systems trained on a single synthetic source. Extension to additional datasets such as SIDTD [**?**] would substantially strengthen the external validity of the conclusions.

b) Architecture.: The DocVerify encoder is a custom lightweight CNN trained from scratch. Transformer-based encoders [**?**] and models pre-trained on large-scale vision datasets typically outperform scratch-trained CNNs when sufficient training data is available. Revisiting the architecture with a stronger encoder is a natural extension, particularly if cross-domain evaluation requires richer feature representations.

c) Challenge evaluation gap.: The comparison in Table V compares our system evaluated on an internal holdout against challenge participants evaluated on the official test set. Although the two sets are drawn from the same distribution (FantasyID), the results are not directly comparable. An official submission to the challenge would eliminate this caveat.

d) Variance on Track 2.: The high variance in localization F1 (± 0.096) across seeds, driven by F1 on bonafide images, represents a practical reliability concern. Future work should investigate targeted training strategies (e.g., calibration of classification confidence to reduce false-positive mask activations on bonafide images).

V. Conclusion

We presented DocVerify, a multi-task deep learning system for joint detection and localization of forged identity documents, and a multi-objective Pareto HPO framework that selects hyperparameter configurations by optimizing detection quality (PR-AUC) and localization quality (Dice) simultaneously, without fixing their relative importance in advance.

Our main findings are:

1) Joint multitask training with Pareto-optimized loss weights achieves significantly higher Dice than both single-task variants and equal-weight joint training (blind test, 30 seeds, $p = 1.3 \times 10^{-4}$, $d = 1.01$), while maintaining competitive PR-AUC.
2) Under a rigorous 10-fold nested cross-validation protocol with 50 MOTPE trials per fold, DocVerify achieves PR-AUC $= 0.9921 \pm 0.0058$ and Dice $= 0.856 \pm 0.030$ on held-out folds, demonstrating consistent generalization across data partitions.
3) Training exclusively on FantasyID and without any external pre-training, DocVerify achieves results competitive with the top-3 entries of the DeepID 2025 Challenge, surpassing the challenge winner on Track 2 localization (0.807 vs. 0.784) and all FantasyID-only systems on Track 1 detection.
4) Multi-objective Pareto HPO is statistically indistinguishable from scalar selection on absolute metrics ($n = 10$, all $p \geq 0.05$), but provides principled, objective-agnostic

model selection and a richer characterization of the HPO objective landscape.

The source code, training scripts, and all experimental artifacts are publicly available at https://github.com/HernanDiaz/IDVerify/.

Several directions remain open. The most pressing is improving cross-domain generalization: models trained on a single synthetic dataset cannot be assumed to transfer automatically to real-world document designs, and evaluating robustness across domains remains future work. Three complementary strategies are worth investigating: (i) *forgery-artifact-centric features* — noise fingerprints, compression inconsistencies, and frequency-domain anomalies [**?**] that are independent of document appearance and thus naturally domain-invariant; (ii) *foundation model backbones* such as DINOv2 or CLIP, whose large-scale pre-training provides richer general-purpose representations that have already shown cross-domain benefits in concurrent work [**?**]; (iii) *domain-adversarial training* [**?**], where a domain discriminator forces the encoder to learn features that are invariant across document types. An official submission to the DeepID 2025 Challenge test server would also provide a directly controlled comparison with published baselines, eliminating the internal-holdout caveat noted in Section IV.

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References

[1] Europol Innovation Lab, "Facing reality? Law enforcement and the challenge of deepfakes," Europol, The Hague, Netherlands, Observatory Report, Apr. 2022. [Online]. Available: https://www.europol.europa.eu/publications-events/publications/facing-reality-law-enforcement-and-challenge-of-deepfakes

[2] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," Official Journal of the European Union, OJ L 2024/1689, 2024, published 12 July 2024; entered into force 1 August 2024. Article 13: Transparency and provision of information to deployers. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng

[3] F. Guillaro, D. Cozzolino, A. Sud, N. Dufour, and L. Verdoliva, "TruFor: Leveraging all-round clues for trustworthy image forgery detection and localization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 19 782–19 792.

[4] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[5] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7482–7491.

[6] P. Korshunov, A. Mohammadi, V. Vidit, C. Ecabert, and S. Marcel, "FantasyID: A dataset for detecting digital manipulations of ID-documents," in *Proceedings of the IEEE International Joint Conference on Biometrics (IJCB)*, Osaka, Japan, 2025. [Online]. Available: https://arxiv.org/abs/2507.20808

[7] M. Huh, A. Liu, A. Owens, and A. A. Efros, "Fighting fake news: Image splice detection via learned self-consistency," in *European Conference on Computer Vision (ECCV)*, 2018, pp. 101–117.

[8] J. H. Bappy, C. Simons, L. Nataraj, B. S. Manjunath, and A. K. Roy-Chowdhury, "Hybrid LSTM and encoder–decoder architecture for detection of image forgeries," vol. 28, no. 7, 2019, pp. 3286–3300.

[9] Y. Wu, W. AbdAlmageed, and P. Natarajan, "ManTra-Net: Manipulation tracing network for detection and localization of image forgeries with anomalous features," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9543–9552.

[10] D. Cozzolino and L. Verdoliva, "Noiseprint: A CNN-based camera model fingerprint," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 144–159, 2020.

[11] K. Triaridis, K. Tsigos, and V. Mezaris, "MMFusion: Combining image forensic filters for visual manipulation detection and localization," in *Proceedings of the 30th International Conference on MultiMedia Modeling (MMM)*, 2024. [Online]. Available: https://arxiv.org/abs/2312.01790

[12] H. Liu, Z. Tan, C. Tan, Y. Wei, Y. Zhao, and J. Wang, "Forgery-aware adaptive transformer for generalizable synthetic image detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 10 770–10 780. [Online]. Available: https://arxiv.org/abs/2312.16649

[13] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 24 480–24 489. [Online]. Available: https://arxiv.org/abs/2302.10174

[14] C. Boned, M. Talarmain, N. Ghanmi, G. Chiron, S. Biswas, A. M. Awal, and O. Ramos Terrades, "Synthetic dataset of ID and travel documents," *Scientific Data*, vol. 11, no. 1, p. 1356, 2024. [Online]. Available: https://www.nature.com/articles/s41597-024-04160-9

[15] J. Muñoz-Haro, R. Tolosana, R. Vera-Rodriguez, A. Morales, and J. Fierrez, "FakeIDet: Exploring patches for privacy-preserving fake ID detection," in *Proceedings of the IEEE International Joint Conference on Biometrics (IJCB)*, 2025. [Online]. Available: https://arxiv.org/abs/2504.07761

[16] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[17] P. Korshunov, V. Vidit, A. Mohammadi, C. Ecabert, N. Shamoska, S. Marcel *et al.*, "DeepID: Challenge of detecting synthetic manipulations in ID documents," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025, pp. 510–519. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025W/DeepID/html/Korshunov_DeepID_Challenge_of_Detecting_Synthetic_Manipulations_in_ID_Documents_ICCVW_2025_paper.html

[18] A. George and S. Marcel, "EdgeDoc: Hybrid CNN-transformer model for accurate forgery detection and localization in ID documents," 2025, idiap Research Report RR-08-2025; 3rd place DeepID 2025 Challenge. [Online]. Available: https://arxiv.org/abs/2508.16284

[19] M. Maaz, A. Shaker, H. Cholakkal, S. Khan, S. W. Zamir, R. M. Anwer, and F. S. Khan, "EdgeNeXt: Efficiently amalgamated CNN-transformer architecture for mobile vision applications," in *European Conference on Computer Vision Workshops (ECCVW)*, 2022, pp. 3–20. [Online]. Available: https://arxiv.org/abs/2206.10589

[20] C. Naseeb, A. A. Cheema, H. Sami, T. Afzal, M. Omair, and U. Habib, "TwoHead-SwinFPN: A unified DL architecture for synthetic manipulation detection and localization in identity documents," 2026. [Online]. Available: https://arxiv.org/abs/2601.12895

[21] Z. Yu, J. Ni, J. Zhang, H. Deng, and Y. Lin, "Reinforced multi-teacher knowledge distillation for efficient general image forgery detection and localization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, oral presentation; base method of Sunlight (1st place, DeepID 2025). [Online]. Available: https://arxiv.org/abs/2504.05224

[22] S. Ruder, "An overview of multi-task learning in deep neural networks," *arXiv preprint arXiv:1706.05098*, 2017.

[23] M. Crawshaw, "Multi-task learning with deep neural networks: A survey," *arXiv preprint arXiv:2009.09796*, 2020.

[24] R. T. Marler and J. S. Arora, "Survey of multi-objective optimization methods for engineering," *Structural and Multidisciplinary Optimization*, vol. 26, no. 6, pp. 369–395, 2004.

[25] S. Liu, E. Johns, and A. J. Davison, "End-to-end multi-task learning with attention," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 1871–1880.

[26] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.

[27] Z. Wu, S. Chen, and X. Zhang, "Robust image forgery detection over online social network shared images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 13 440–13 449.

[28] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 24, 2011.

[29] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," vol. 13, 2012, pp. 281–305.

[30] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 2623–2631.

[31] K. Miettinen, *Nonlinear Multiobjective Optimization*. Kluwer Academic Publishers, 1999.

[32] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.

[33] S. Watanabe and F. Hutter, "Multiobjective tree-structured parzen estimator," vol. 73, 2022, pp. 1–31.

[34] Y. Patel, S. Tanwar, P. Bhattacharya, R. Gupta, T. Alsuwian, I. E. Davidson, and T. F. Mazibuko, "An improved dense CNN architecture for deepfake image detection," *IEEE Access*, vol. 11, pp. 22 081–22 095, 2023, open access, CC BY-NC-ND 4.0.

[35] B. Xu, N. Wang, T. Chen, and M. Li, "Empirical evaluation of rectified activations in convolutional network," *arXiv preprint arXiv:1505.00853*, 2015.

[36] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015, pp. 448–456.

[37] L. R. Dice, "Measures of the amount of ecologic association between species," *Ecology*, vol. 26, no. 3, pp. 297–302, 1945.

[38] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proceedings of the 4th International Conference on 3D Vision (3DV)*, 2016, pp. 565–571.

[39] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[40] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, "Mixed precision training," in *International Conference on Learning Representations (ICLR)*, 2018.

[41] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," vol. 32, 2019.

[42] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, no. 1, p. 91, 2006.

[43] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Lawrence Erlbaum Associates, 1988.

[44] D. Cozzolino and L. Verdoliva, "Noiseprint: A CNN-based camera model fingerprint," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 144–159, 2020.

[45] Y. Ganin, E. Ustunova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016.